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FACULTY OF INFORMATION TECHNOLOGY

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FINE-TUNING DEEPSEEK-OCR WITH VIETNAMESE DATASET

Course: Introduction to Natural Language Processing

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Mục lục

1 Background 2

1.1 Overview of OCR and Challenges in Vietnamese 2

1.2 Overview of the DeepSeek-OCR Model 2

2 Methodology 3

2.1 Detailed Description of the Pipeline 3

2.1.1 Data Collection and Preprocessing 3

2.1.2 Training Environment Setup 3

2.1.3 Fine-tuning Details 4

2.2 Inference and Post-processing 5

3 Experiments 5

3.1 Sampling Strategy 5

3.2 Dataset Splits 5

3.3 Experimental Design: Baseline vs. Fine-tuned Model 6

3.4 Evaluation Metric 6

4 Results 7

4.1 Quantitative Results 7

4.2 Quantitative Analysis 7

4.3 Qualitative Analysis 7

4.4 Illustrative Output Examples 8

5 Discussion 8

5.1 Assessment of Performance Improvement 8

5.2 Reasons for Performance Improvement and Remaining Errors 8

5.2.1 Reasons for Improvement 8

5.2.2 Reasons for Limited or Remaining Errors 9

5.3 Limitations of the Dataset, Model, and Training Process 9

5.3.1 Dataset Limitations 9

5.3.2 Model Limitations 9

5.3.3 Training Process Limitations 10

6 Conclusion 10

1 Background

1.1 Overview of OCR and Challenges in Vietnamese

Optical Character Recognition (OCR) is a fundamental technology in natural language processing that enables the conversion of text embedded in images (e.g., scanned documents or handwritten text) into machine-readable, editable, and searchable textual data. While OCR systems have achieved high accuracy in several major languages, Vietnamese OCR—especially for handwritten text — remains a challenging task due to language-specific characteristics.

One major challenge lies in the **complex diacritic system** of Vietnamese. The language employs multiple tonal marks (acute, grave, hook above, tilde, and dot below) as well as modified characters with additional marks (e.g., *â, ã, ê, ô, ư*, and *đ*). In handwritten text, these diacritics are often small, inconsistently positioned, merged with neighboring characters, or occasionally omitted, which significantly complicates character-level recognition.

Another critical difficulty is the **high variability of handwriting styles**. Handwritten Vietnamese text exhibits substantial diversity in stroke shapes, character sizes, slant angles, and degrees of cursiveness. This variability poses significant challenges for OCR models in terms of generalization, particularly when encountering writing styles that differ from those seen during training.

1.2 Overview of the DeepSeek-OCR Model

DeepSeek-OCR is an advanced multimodal Vision–Language Model (VLM) designed to jointly process visual and textual information for document understanding and OCR-related tasks.

Architecture: The model adopts a hybrid architecture that integrates a vision encoder and a large language model (LLM) decoder. The vision encoder is responsible for extracting high-level visual features from input images, while the LLM decoder generates coherent textual sequences based on both the extracted visual representations and the textual prompt. This design enables DeepSeek-OCR to handle complex document analysis tasks that require both visual perception and linguistic reasoning.

Input/Output:

- **Input:** An image (e.g., .jpg or .png, typically represented as a PIL Image object) combined with a textual prompt (e.g., `<image>\n Vietnamese OCR`).
- **Output:** A textual sequence corresponding to the content recognized from the input image, generated as a response to the provided prompt.

Limitations of the Pre-trained Model: Despite its strong general capabilities, the pre-trained DeepSeek-OCR model is primarily optimized for high-resource languages such as English and Chinese. As a result, when applied directly to Vietnamese—particularly handwritten Vietnamese text - the model often exhibits suboptimal performance. Common errors include incorrect recognition of tonal marks, confusion between visually similar characters, and poor handling of Vietnamese-specific handwriting patterns.

2 Methodology

2.1 Detailed Description of the Pipeline

The proposed methodology consists of three main stages: data preparation, training environment setup, and model fine-tuning.

2.1.1 Data Collection and Preprocessing

- **Data collection:** The experiments are conducted using the UIT-HWDB-line (Handwritten Vietnamese Database), which is organized into two primary directories: `train_data` and `test_data`. Each directory contains multiple numbered subfolders, where each subfolder includes a set of handwritten image files and a corresponding `label.json` file storing the ground-truth textual annotations.
- **Preprocessing:** The raw data are processed through the following steps:
 - Parsing the `label.json` files to extract image paths and corresponding text labels.
 - Converting all images to RGB format to ensure consistency across inputs.
 - Transforming the dataset into the `ImageFolder` format supported by the HuggingFace `datasets` library, consisting of an image directory and a `metadata.jsonl` file that maps image paths to text labels.
 - Splitting the data into training and validation sets using a 90%/10% ratio derived from `train_data`, while preserving the original `test_data` as an independent test set for final evaluation.

2.1.2 Training Environment Setup

- **Hardware:** All experiments are conducted on Google Colab using an NVIDIA T4 GPU with 16GB of VRAM.
- **Software libraries:** The training pipeline leverages several specialized libraries: `unsloth` for memory-efficient loading and fine-tuning of large models, `transformers` for model architecture and training utilities, `peft` for parameter-efficient fine-tuning with LoRA, `trl` for the training framework, and `jiwer` for computing the Character Error Rate (CER).
- **Memory optimization:** The DeepSeek-OCR model is loaded using 4-bit quantization via Unsloth, significantly reducing GPU memory consumption and enabling training on limited hardware.

2.1.3 Fine-tuning Details

```
... =====
BASE: torch.Size([1, 256, 1280])
PATCHES: torch.Size([9, 100, 1280])
=====
[90/90 2:24:03, Epoch 0/1]

Step Training Loss Validation Loss
30      1.019800      0.874294
60      0.664900      0.785732
90      0.905400      0.750494

=====
BASE: torch.Size([1, 256, 1280])
PATCHES: torch.Size([9, 100, 1280])
=====
```

- **Fine-tuning strategy:** The model is fine-tuned using Quantized Low-Rank Adaptation (QLoRA). This approach allows efficient adaptation of large-scale models by training only a small number of additional parameters (LoRA adapter matrices) while keeping the original model weights frozen, making it suitable for resource-constrained environments.
- **LoRA configuration:** The LoRA parameters are defined as follows:
 - Rank: $r = 16$
 - Scaling factor: $\alpha = 16$
 - Dropout: 0
 - Bias: none
 - Target modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`

These target modules correspond to the attention and feed-forward components of the Transformer architecture.

- **Training hyperparameters:** The training process employs the following settings:
 - Per-device training batch size: 1
 - Gradient accumulation steps: 8 (effective batch size of 8)
 - Per-device evaluation batch size: 1
 - Warmup steps: 10
 - Maximum training steps: 90
 - Learning rate: 2×10^{-4}
 - Logging frequency: every step
 - Optimizer: 8-bit AdamW
 - Weight decay: 0.001

- Learning rate scheduler: linear
- Random seed: 3407
- Checkpoint saving strategy: every 30 steps
- Evaluation strategy: every 30 steps
- Mixed precision: FP16 (BF16 not supported on NVIDIA T4)
- Removal of unused columns: disabled
- Reporting: disabled
- Best model selection: enabled based on validation performance

2.2 Inference and Post-processing

- **Inference procedure:** A custom inference function, `predict_manual`, is employed instead of the default `model.infer` method. This function handles image preprocessing (RGB conversion, dynamic cropping, and padding), constructs image placeholder tokens, tokenizes the textual prompt, and generates the output sequence using `model.generate`.
- **Post-processing:** The generated token sequences are decoded into plain text by removing special tokens (`skip_special_tokens=True`). No additional post-processing is applied beyond trimming redundant whitespace.

3 Experiments

3.1 Sampling Strategy

Due to the computational constraints of the Google Colab environment with an NVIDIA T4 GPU, a sampling strategy was applied to the UIT-HWDB-line dataset to ensure feasible training and evaluation.

- **Training and validation data:** The training and validation sets are constructed from the first 249 subdirectories of the `train_data` split in the UIT-HWDB-line dataset.
- **Independent test data:** The test set is built from 6 subdirectories of the original `test_data` split. This test set is strictly independent and has no overlap with the training or validation data, ensuring an unbiased final evaluation.

3.2 Dataset Splits

The dataset is partitioned into three subsets with the following proportions and purposes:

- **Training set:** Approximately 90% of the samples from the 249 selected subdirectories of `train_data`. This set which has 6325 images used to optimize the model parameters during fine-tuning.
- **Validation set:** The remaining 10% of the samples from the same 249 subdirectories. This set which has 703 images is employed to monitor training progress, tune hyperparameters, and select the best-performing model checkpoint.

- ```
 Starting Data Processing...
-> Scanning 249 folders in train_data...

Reading Folders: 100%  249/249 [02:29<00:00, 1.80it/s]

-> Scanning 6 folders in test_data...

Reading Folders: 100%  6/6 [00:04<00:00, 1.72it/s]

-> Saving train (6325 images)...

Writing train: 100%  6325/6325 [58:14<00:00, 1.54it/s]

-> Saving validation (703 images)...

Writing validation: 100%  703/703 [06:28<00:00, 1.78it/s]

-> Saving test (201 images)...

Writing test: 100%  201/201 [01:46<00:00, 1.79it/s]

 Data processing complete. Saved to: /content/drive/MyDrive/NLP/data
```

The experiments are designed to directly compare the performance of the pre-trained DeepSeek-OCR model and its fine-tuned counterpart under identical evaluation conditions:

- CER is defined as the normalized Levenshtein distance between the predicted character sequence and the ground-truth sequence. Formally, let  $S$  denote the number of character substitutions,  $D$  the number of deletions,  $I$  the number of insertions required to transform the predicted sequence into the reference sequence, and  $N$  the total number of characters in the ground-truth sequence. The Character Error Rate is computed as:

Page 6

## 4 Results

### 4.1 Quantitative Results

Table 1 reports the average Character Error Rate (CER) achieved by the original pre-trained DeepSeek-OCR model and the fine-tuned model on the independent test set.

Bảng 1: Average CER on the independent test set

| Model                              | Average CER |
|------------------------------------|-------------|
| DeepSeek-OCR (Original – Baseline) | 0.3040      |
| DeepSeek-OCR (Fine-tuned)          | 0.2002      |

### 4.2 Quantitative Analysis

The results demonstrate a substantial performance improvement after fine-tuning the DeepSeek-OCR model on handwritten Vietnamese data. The average CER decreases from  $CER_{base}$  for the baseline model to  $CER_{ft}$  for the fine-tuned model.

The absolute improvement in CER is defined as:

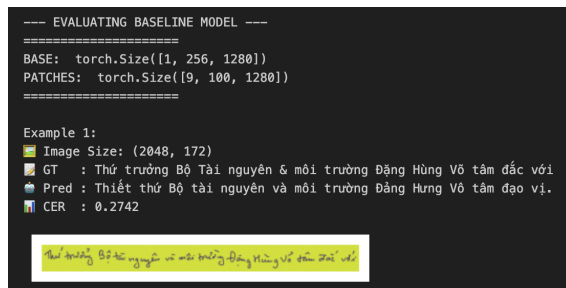
$$\Delta CER = CER_{base} - CER_{ft} = 0.1038 \quad (2)$$

This reduction in CER indicates that the fine-tuning process significantly enhances character-level recognition accuracy. In particular, the improvement confirms the effectiveness of LoRA-based adaptation in transferring a general-purpose OCR model to a low-resource, diacritic-rich language such as Vietnamese.

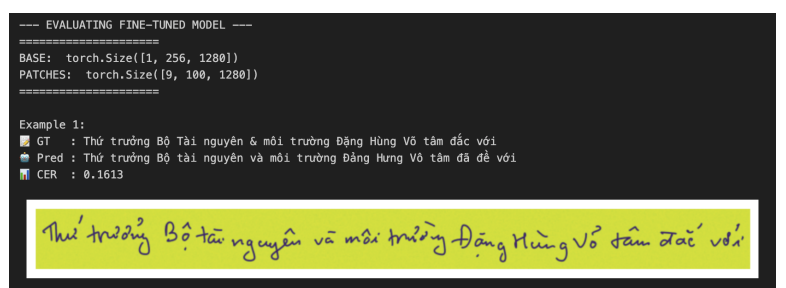
### 4.3 Qualitative Analysis

Beyond quantitative metrics, qualitative inspection of model predictions reveals several notable improvements.

**Improved recognition of Vietnamese diacritics:** The baseline model frequently omits or misclassifies tonal marks and diacritical symbols, which leads to semantic distortion in Vietnamese text. In contrast, the fine-tuned model demonstrates a much stronger ability to correctly identify and place tonal marks and modified characters.

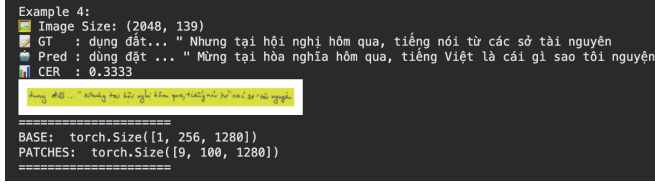


(a) Original DeepSeek-OCR

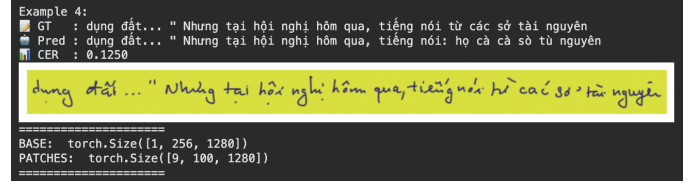


(b) Fine-tuned DeepSeek-OCR

**Better handling of handwritten variations:** The fine-tuned model is more robust to non-standard handwriting styles, including cursive strokes, merged characters, and inconsistent spacing. This results in fewer meaningless characters, reduced character fragmentation, and improved word integrity.



(a) Original DeepSeek-OCR



(b) Fine-tuned DeepSeek-OCR

## 4.4 Illustrative Output Examples

Figure-based qualitative examples from the test set further highlight the contrast between the baseline and fine-tuned models. The baseline model often removes diacritics entirely or merges words incorrectly, while the fine-tuned model preserves both lexical content and diacritical accuracy.

These qualitative improvements align closely with the observed reduction in CER, confirming that the fine-tuned model not only performs better numerically but also produces outputs that are linguistically and semantically more faithful to handwritten Vietnamese text.

## 5 Discussion

### 5.1 Assessment of Performance Improvement

The observed reduction in Character Error Rate (CER), from 0.3040 to 0.2002, represents a highly encouraging outcome. This substantial decrease in error rate demonstrates that fine-tuning the DeepSeek-OCR model with domain-specific data from UIT-HWDB-line is an effective strategy for adapting a large Vision-Language Model to the particular challenges of handwritten Vietnamese OCR.

Importantly, this improvement is achieved under constrained computational resources. The use of QLoRA enables efficient parameter adaptation on a single NVIDIA T4 GPU within the Google Colab environment, highlighting the practicality of parameter-efficient fine-tuning for low-resource research and deployment settings.

### 5.2 Reasons for Performance Improvement and Remaining Errors

#### 5.2.1 Reasons for Improvement

Several factors contribute to the observed performance gains:

- **Learning language-specific characteristics:** Through fine-tuning, the model acquires a better understanding of Vietnamese character distributions and orthographic patterns, particularly the interaction between base characters and tonal diacritics, which is essential for semantic correctness.

- **Adaptation to handwritten styles:** The UIT-HWDB-line dataset exposes the model to a wide range of Vietnamese handwriting variations, including cursive writing, stroke merging, and inconsistent spacing. This exposure allows the fine-tuned model to better distinguish meaningful character patterns from noise that often confuses the baseline model.
- **Domain adaptation:** Fine-tuning effectively specializes the model from a general-purpose OCR system into a domain-specific model tailored for handwritten Vietnamese text, leading to more accurate and linguistically coherent predictions.

### 5.2.2 Reasons for Limited or Remaining Errors

Despite the improvements, some limitations remain:

- **Limited training duration:** The number of training steps (`max_steps = 90`) is relatively small. Additional training iterations could further refine the model's representations and reduce residual errors.
- **Training data scale:** Although 249 subdirectories are used for training and validation, this subset may still be insufficient to fully capture the extensive diversity of handwritten Vietnamese text.
- **Image quality constraints:** The model continues to struggle with extremely low-resolution images, severe blurring, or highly merged strokes, where critical visual cues are partially or entirely lost.

## 5.3 Limitations of the Dataset, Model, and Training Process

### 5.3.1 Dataset Limitations

- **Sample size:** The use of a sampled subset of the UIT-HWDB-line dataset restricts the range of handwriting styles available during training, potentially limiting the model's ability to generalize.
- **Coverage:** While UIT-HWDB-line is a valuable resource, it may not fully represent all real-world Vietnamese handwriting styles encountered in practical applications.

### 5.3.2 Model Limitations

- **Architectural constraints:** Although DeepSeek-OCR is a powerful Vision–Language Model, its architecture may have inherent limitations when dealing with extreme handwriting irregularities.
- **Generalization ability.** The fine-tuned model may experience performance degradation when applied to handwriting styles or writing instruments that fall outside the distribution of the UIT-HWDB-line dataset.

### 5.3.3 Training Process Limitations

- **Computational resources:** The constraints of the Google Colab T4 GPU (16GB VRAM) necessitate small batch sizes and limited training steps, which may prevent full convergence. Due to the absence of additional paid computing credits, the experiments were conducted solely within the time limits of the free Google Colab environment. As a result, model training on the NVIDIA T4 GPU was constrained to the default session duration provided by Colab, which limited the total number of training steps.
- **LoRA hyperparameter selection:** The chosen LoRA parameters (e.g., rank  $r$  and scaling factor  $\alpha$ ) may not be globally optimal, and further hyperparameter exploration could yield additional performance gains.

## 6 Conclusion

This project successfully fine-tuned the DeepSeek-OCR model using a LoRA-based approach to significantly improve the recognition of handwritten Vietnamese text. Experimental results demonstrate a clear reduction in Character Error Rate (CER) compared to the original baseline model, with particularly notable improvements in the recognition of Vietnamese diacritics and handwriting-specific characteristics.

These findings highlight the strong potential of large Vision–Language Models for addressing language-specific OCR challenges when appropriately adapted to the target domain. Moreover, the results confirm the effectiveness of resource-efficient fine-tuning techniques such as QLoRA, which enable meaningful performance gains even under limited computational constraints.

Overall, this work provides empirical evidence that parameter-efficient adaptation of large OCR models is a promising direction for developing more accurate and practical Vietnamese handwritten OCR systems, and it lays a solid foundation for future research and real-world applications in this domain.

## Code Availability

The full implementation, including data preprocessing scripts, fine-tuning configuration, and evaluation code, is publicly available at: [https://github.com/DanielNguyen-05/DeepSeek-OCR\\_Vietnamese](https://github.com/DanielNguyen-05/DeepSeek-OCR_Vietnamese)